Simplex Tableau — Summarizing Information

min: $c^T x$

s.t. $Ax = b$

$x \geq 0$

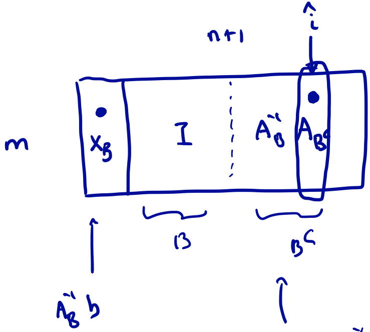
"$Ax = b$" $\longleftrightarrow$ "$A_B^{-1} A x = A_B^{-1} b$"

$$A = \left[A_B \; \vdots \; A_{B^c} \right]$$

$$A_B^{-1} A_B = [\underbrace{I}_{\text{location of } I \text{ encodes } B} \; \vdots \; \underbrace{A_B^{-1} A_{B^c}}_{}]$$

encodes all possible d_B vectors for any choice $\hat{i}$ & B

Basic Tableau



$x_{new} = x + \theta d$

$d_B = - A_B^{-1} A_{\hat{i}}$

$d_B = A_B^{-1} A_{\hat{i}}, \quad \hat{i} \notin B$

Choose $\hat{i} \notin B$.

θ: ratio with elts of x_B

$= \min_{j : d_j > 0} \frac{x_j}{d_j}$: $\hat{j}$ exits the basis.

$B_{new} = (B \setminus \hat{j}) \cup \hat{i}$

An Example — 1/3

$$\begin{aligned}
3x_1 + 2x_2 + x_3 \phantom{{}+x_4+x_5} &= 1 \\
5x_1 + x_2 + x_3 + x_4 \phantom{{}+x_5} &= 3 \\
2x_1 + 5x_2 + x_3 \phantom{{}+x_4} + x_5 &= 4
\end{aligned}$$

$$\underbrace{}_{b}$$

$$\left[\begin{array}{c|c} b & A \end{array}\right]$$

$$\left[\begin{array}{c|ccccc}
1 & 3 & 2 & 1 & 0 & 0 \\
3 & 5 & 1 & 1 & 1 & 0 \\
4 & 2 & 5 & 1 & 0 & 1
\end{array}\right]$$

← subtract 1x row 1 from row 2 & row 3

$$\left[\begin{array}{c|ccccc}
1 & 3 & 2 & 1 & 0 & 0 \\
2 & 2 & -1 & 0 & 1 & 0 \\
3 & -1 & 3 & 0 & 0 & 1
\end{array}\right]$$

$A_B^{-1} b$ $\qquad$ $A_B^{-1} A_{B^c}$ $\qquad$ $B = \{3, 4, 5\}$

$$x = \begin{pmatrix} x_B \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}$$ is an BFS

$x_B = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

An Example — 2/3

$$\left[\begin{array}{c|ccccc} 1 & 3 & 2 & 1 & 0 & 0 \\ 2 & 2 & -1 & 0 & 1 & 0 \\ 3 & -1 & 3 & 0 & 0 & 1 \end{array}\right]$$

$$\underbrace{\quad\quad\quad}_{B^c}$$

$B = \{3, 4, 5\}.$

Choose $\hat{i} = 1.$

$A_B^{-1} A_1$

$$x_{new} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \\ 3 \end{pmatrix} + \theta \begin{pmatrix} 1 \\ 0 \\ -3 \\ -2 \\ +1 \end{pmatrix}$$

$$\theta = \min\left\{ \frac{1}{3}, \frac{2}{2} \right\}$$

pivot element

$$\boxed{\hat{j} = 3.}$$

An Example — 3/3

pivot element

$$\begin{array}{c|ccccc} & \hat{i} & & & & \\ \hline 1 & 3 & 2 & 1 & 0 & 0 \\ 2 & 2 & -1 & 0 & 1 & 0 \\ 3 & -1 & 3 & 0 & 0 & 1 \end{array}$$

must look like $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\hat{i}$

$\hat{j} = 3$ exits basis.

$B_{new} = \{1, 4, 5\}$

Divide 1^{st} row by 3 (pivot elt) and subtract it $\times 2$ from 2^{nd} row add it to 3^{rd} row.

$$\begin{array}{c|ccccc} \frac{1}{3} & 1 & \frac{2}{3} & \frac{1}{3} & 0 & 0 \\ \frac{4}{3} & 0 & -\frac{7}{3} & -\frac{4}{3} & 1 & 0 \\ \frac{10}{3} & 0 & \frac{11}{3} & \frac{1}{3} & 0 & 1 \end{array}$$

$B = \{1, 4, 5\}$ $\quad x_B = \begin{pmatrix} 1/3 \\ 0 \\ 0 \\ 4/3 \\ 10/3 \end{pmatrix}$

Recall Reduced Costs

$$\bar{c} = c - \underbrace{c_B^T A_B^{-1}}_{p} A \quad : \quad \bar{c} = c - p^T A$$

$$p = c_B^T A_B^{-1}.$$

if $i \in B$: $\bar{c}_i = 0$.

if $\bar{c} \geq 0 \Rightarrow B$ is $\begin{pmatrix} x_B \\ 0 \end{pmatrix}$ optimal

Tracking Reduced Costs — 1/3

Add 0^{th} row to tableau:

$$
\begin{array}{c|c|c|}
 & & n+1 \\
 & -c^T x \;\; (-c_B^T A_B^{-1} b) & c - c_B^T A_B^{-1} A \\
\hline
m+1 & A_B^{-1} b & A_B^{-1} A
\end{array}
$$

$$\left[\, I \;\middle|\; A_B^{-1} A_{B^c} \,\right]$$

How to update zeroth row?

$\bar{c}_i = 0$ for $i \in B$.

Need to make $\bar{c}_{\hat{i}} = 0$

<u>Claim:</u> If we make $\bar{c}_{\hat{i}} = 0$ by adding multiple of pivot row, then 0^{th} row becomes:

$$
\begin{array}{|c|c|}
\hline
-c_{B_{new}}^T A_{B_{new}}^{-1} b & c - c_{B_n}^T A_{B_n}^{-1} A \\
\hline
\end{array}
$$

Tracking Reduced Costs — 2/3

Proof of this claim:

$$[0 \mid c] - c_B^T A_B^{-1} [b \mid A]$$

add multiple of pivot row: row $\hat{j}$ corresponding to exiting basis variable.

$\underline{h^T [b \mid A]}$, h is $\hat{j}^{\text{th}}$ row of A_B^{-1}.

$$[0 \mid c] - \underbrace{(c_B^T A_B^{-1} + \alpha h^T)}_{p} [b \mid A]$$

$\underline{p}$ ⟵ what is p?

Tracking Reduced Costs — 3/3

α chosen to make

$\bar{c}_{\hat{i}} = 0$: $\quad \bar{c}_{\hat{i}} = c_{\hat{i}} - p^T A_{\hat{i}} = 0.$

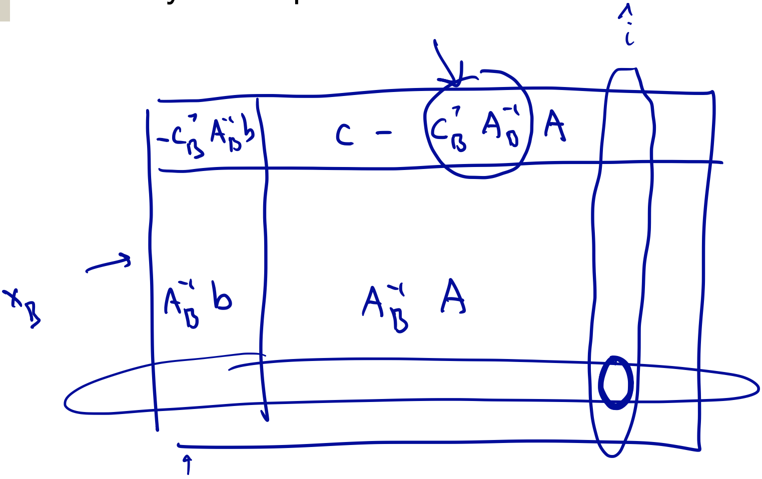
Also: $\quad c_j - p^T A_j = 0 \quad \forall j \in B \setminus \hat{j}$

$c_i - p^T A_i = 0 \quad \forall i \in (B \setminus \hat{j}) \cup \hat{i} = B_{new}.$

$c_{B_{new}} - p^T A_{B_{new}} = 0$:

$$p = c_{B_{new}}^T A_{B_{new}}^{-1}$$

Summary of Simplex via Tableau



1. Choose $\hat{i}$ s.t. $\bar{c}_{\hat{i}} < 0$
2. Find pivot elt.
3. Add multiples of pivot row to make pivot column: $\begin{bmatrix} \boxed{0} \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

Example continued.

pivot row

0	-2	-1	0	0	0
1	3	2	1	0	0
2	2	-1	0	1	0
3	-1	3	0	0	1

$-d_B = A_B^{-1} A_1$

min: $c_1 x_1 + c_2 x_2 + \underbrace{0 x_3 + 0 x_4 + 0 x_5}$

2/3	0	1/3	2/3	0	0
1/3	1	2/3	1/3	0	0
4/3	0	-7/3	-4/3	1	0
10/3	0	11/3	1/3	0	1

Now at opt sol'n

as $\bar{c} \geq 0$.

Three row operations :

① B_{new}

② $A_{B_n} = \begin{bmatrix} 3 & 0 & 0 \\ 5 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

③ Compute $A_{B_n}^{-1}$

④ Compute: $A_{B_n}^{-1} b$, $A_{B_n}^{-1} A$,

$$\bar{c} = c - c_{B_n}^{T} A_{B_n}^{-1} A$$

i

0	-2	-1	0	0	0
0	3	2	1	0	0
2	2	-1	0	1	0
3	-1	3	0	0	1

$B = \{3, 4, 5\}$

$x = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 2 \\ 3 \end{pmatrix}$

$\theta = 0$

0	0	1/3	2/3	0	0
0	1	2/3	1/3	0	0
2	0	-7/3			
3	0				

same

$B = \{1, 4, 5\}$

$x = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 2 \\ 3 \end{pmatrix}$